UTM
UNIVERSITI TEKNOLOGI MALAYSIA

COURSE INFOR Clear Form Copy Sheet Update Index

School/Faculty:	**Faculty of Computing**		
Program name:	**Bachelor in Software Engineering with Honours**		
Course code:	**SCSE 3143**	**Academic Session/Semester:**	**2026 2027 1**
Course name:	**Ubiquitous Computing**	**Pre/co requisite (course name and code, if applicable):**	**SCSR 2143 Operating System**
Credit hours:	**3**		

Course synopsis	The goal of this course is to acquaint the fundamental concepts and state-of-the-art in Ubiquitous Computing technology, which is also known as Pervasive Computing. Ubiquitous Computing is the result of computer technology advancing that is, computing devices are becoming progressively smaller, more powerful and more connected and this has brought with it a trend toward artifacts having hardware and software embedded within them. These devices can be embedded into cars, airplanes, ships, bikes, posters, signboards, walls and even clothes. This course focuses on the understanding elements involved in designing and building Internet of Things and embedded systems-based Environments. It thus covers independent information devices including but not limited to wearable computers, mobile phones, smart phones, smart-cards, wireless sensor compute nodes etc. and the services made available by them in typical everywhere computing environments. It includes select aspects of human-computer interaction and real-time software design.
Course coordinator (if applicable)	**Dr Muhammad Khatibsyarbini Abd Rahim**

Course lecturer(s)	**Name**	**Office**	**Contact no.**	**E-mail**
	Dr Muhammad Khatibsyarbini Abd Rahim	**N28-402-09**		**khatibsyarbini@utm.my**
	Prof Dr. Dayang Norhayati Abang Jawawi	**N28-305-03**		**dayang@utm.my**

Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	CLO Statements	PLO (MQF Cluster Code)	Taxonomies & Graduate attributes	ESD	VBE	T&L methods	Assessment methods
CLO1	Solve the ubiquitous problems in developing Internet of Things and embedded software system.	PLO2 (CG) C2	C3 (SC2)			Tutorial, Active Learning	Problem Solving (15%), Mid-Term Test (25%), Quiz(10%)
CLO2	Demonstrate the use of appropriate methods and tools to solve complex, integrated problems in developing ubiquitous computing software.	PLO3 (PS) C3A	P4 (SC3)	IPS		Project Based Learning	Project Assignment (20%), Project Development (20%)
CLO3	Demonstrate an entrepreneurial mindset to achieve excellence in managing ubiquitous computing projects across multi-disciplinary environments	PLO10 (ENT) C4B	A5 (ES5)		Excellence	Project Based Learning	Project Proposal (10%)

This is the basic mapping required for the CI. Any added information is allowed (extra columns for weight or other elements) provided this is made consistent for all CI at program/school/faculty level.

1Up to 5 CLO

2MQF Cluster Code
C1 = Knowledge & Understanding, **C2** = Cognitive Skills, **C3A** = Practical Skills, **C3B** = Interpersonal Skills, **C3C**= Communication Skills, **C3D** = Digital Skills, **C3E** = Numercy Skills, **C3F** = Leadership, Autonomy & Responsibility, **C4A** = Personal Skills, **C4B** = Enterpreneurial Skills, **C5** = Ethics & Professionalism

3Taxonomies of Learning Domain and **4UTM's Graduate Attributes** for measurement of outcomes achievement

5Education for Sustainable Development (ESD)
Systems Thinking (ST); Anticipatory Thinking (AT); Normative (N); Strategic Thinking (STT); Collaboration (C); Critical Thinking (CT); Self-Awareness (SA; Integrated Problem Solving (IPS)

7Value-Based Education (VBE)
Integrity; Synergy; Excellence; Sustainability

7T – Test; Q – Quiz; HW – Homework; Asg – Assignment; PR – Project; Pr – Presentation; F – Final Exam etc. Please specify only the related type of assessment.

Details on Innovative T&L practices:

No.	Type	Implementation
1	Active learning	Conducted through in-class activities
2	Project-based learning	Conducted through design assignments. Students in a group of 4 are given two projects related to the real-time requirement, design, development involving the knowledge to adapt and apply known software engineering solution to real-time requirements.
3		

Transferable skills (generic skills learning in course of study which can be useful and utilised in other settings) :

Student learning time (SLT) / Effective Learning Time (ELT) details:

Week/ Meeting	Course Content Outline and Subtopics	CLO* Compulsory	Learning and Teaching Activities										TOTAL SLT
			Face-to-Face (F2F)								Non F2F Independent Learning		
			Physical				Online (Synchronous)				Online (Asynchronous)	Others	
			L	T	P	O	L	T	P	O			
Week 1	Introductory Concepts	CLO1					2			1	2	2	7
Week 2	Internet of Things (IoT)	CLO1	2			1					2	2	7
Week 3	Embedded Real-Time Systems	CLO1	2		1						2	2	7
Week 4 Deepavali | 8 & 9 November 2026 (Sunday & Monday)	Concurrent Programming & RTOS	CLO2					1		1	1	2	2	7
Week 5	Concurrent Programming & RTOS	CLO2					1		1	1	2	2	7

Week 6	Real-Time Analysis Group Assignment	CLO1	2		1					2	2	7
Week 7	Real-Time Analysis Mid-Term	CLO1	2		1					2	2	7
Week 8	MID-SEM BREAK											
Week 9	Context - Aware Computing Submission Group Assignment	CLO1	2			1				2	2	7
Week 10	Wearable & Mobile Computing Quiz 1	CLO2	2		1					2	2	7
Week 11 Christmas Day | 25 December 2026 (Friday)	Human - Computer Interaction Quiz 2	CLO3					1		1	2	2	7
Week 12	Project Development Project Proposal	CLO2					1		1	2	2	7
Week 13	Project Development Submission Project Proposal	CLO2			2	1				2	2	7
Week 14	Project Development	CLO2			2	1				2	2	7
Week 15	Demo Submission Full Project Report	CLO3			2	1				2	2	7
										SUB-TOTAL SLT :		**98**

Continous Assessment		**%**	**Face-to-Face (F2F)**		**NF2F Independent Learning for Assessment**		**TOTAL SLT**
			Physical	**Online (Synchronous)**	**Online (Asynchronous)**	**Others**	
1	Quiz (2)	10	1	1	1	1	4
2	Problem Solving (3)	15	1	1	1	1	4
3	Assignment	20	1	1	1	1	4
4	Mid-Term Test	25	2				2
							0
Assessment activities carried out during the semester to monitor student progress (e.g., quizzes, assignments, projects).					**SUB-TOTAL SLT :**		**14**

Final Assessment	**%**	**Face-to-Face (F2F)**		**NF2F Independent Learning for Assessment**		**TOTAL SLT**
		Physical	**Online (Synchronous)**	**Online (Asynchronous)**	**Others**	

1	[AA] Project Development	20	1	1	1	1	4
2	[AA] Project Proposal	10	1	1	1	1	4
3							0
4							0
5							0

Assessment conducted at the end of the course to evaluate the achievement of overall learning outcomes (e.g., final exam, final project).

SUB-TOTAL SLT :	**8**
SLT for Assessment:	**22**
GRAND TOTAL SLT:	**120**

A	% SLT for F2F Physical Component	28.33
B	% SLT for Online & Independent Learning Component :	71.67
C	%SLT for Online Component:	44.17
D	% SLT for All Practical Component:	11.67
D1	% SLT for F2F Physical Practical Component:	8.33
D2	% SLT for F2F Online Practical Component:	3.33
Please tick (/) if this course is Industrial Training/ Clinical Placement/ Practicum using 50% of Effective Learning Time (ELT)		

Identify special requirement or resources to deliver the course (e.g.,software,nursery, computer lab, simulation room etc)

Computer lab with Java, RTOS and IOT equipments.

References (include required and further readings, and should be the most current)

Text book (if applicable)
Note: This course should refer mainly to research papers like published journals and conference proceedings that describe current issues and solutions in real-time software engineering and software development in general, and specifically in each phase of software development.

Main references

1. Cooling J. (2003). “Software Engineering for Real-Time Systems”, Addison-Wesley.
2. Minh Duc B. (2007). “Real-time object uniform design methodology with UML”, Netherlands: Springer.
3. Khan, M.A., Saeed, S., Darwish, A., Abraham, A. (2014), “Embedded and Real Time System Development: A Software Engineering Perspective Concepts, Methods and Principles”, Springer.
4. Ben-Ari M., Principles of Concurrent and Distributed Programming, Addison-Wesley, England: 2006.
5. Krumm, J. (2009). Ubiquitous Computing Fundamentals. CRC Press.

Learning Management System (LMS)
http://elearning.utm.my

Other additional information (if applicable)

Academic honesty and plagiarism: *(Below is just a sample)*

Assignments are individual tasks and NOT group activities (UNLESS EXPLICITLY INDICATED AS GROUP ACTIVITIES)
Copying of work (texts, simulation results etc.) from other students/groups or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such. Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of zero for the assignment and disciplinary steps may be taken by the Faculty. It is also unacceptable to do somebody else’s work, to lend your work to them or to make your work available to them to copy.

Other additional information (if applicable)

1. To pass this course, students must get at least minimum pass marks following UTM regulation (40%).

2. Attendance is compulsory and will be taken in every lecture session. Student with less than 80% of total attendance is not allowed to sit for final exam.

3. Students are required to behave and follow the University's dressing regulation and etiquette all the time.

4. Exercises and tutorial will be given in class and some may be taken for assessment. Students who do not do the exercise will lose the coursework marks for the exercise.

5. Assignments must be submitted on the due dates. Some points will be deducted for late submissions. Assignments submitted three days after the due date will not be accepted.

6. Make up exam will not be given, except to students who are sick and submit medical certificate confirmed by UTM panel doctors. Make up e

Disclaimer:

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While every effort has been made to ensure the accuracy of the information supplied herein, Universiti Teknologi Malaysia cannot be held responsible for any errors or omissions.

ELT = (Theory + Industrial Guidance + Assessment) x 50%

Total of credit for LI/Practical = ELT/40 Notional Hours
Note: For ODL Programme : Courses with mandatory practical requirement imposed by programme standards or any related standards can be exempted from complying to the minimum 80% ODL delivery rule in the SLT.

Prepared by:		**Certified by:**	
Name:	Dr. Muhammad Khatibsyarbini	**Name:**	
Signature:			
		Signature:	
Date:	30-Aug-26	**Date:**	30-Aug-26

Compulsory

Example: This course introduces the fundamental concepts and principles of human resource development (HRD), focusing on both theoretical and practical aspects. Students will explore topics such as employee training, career development, performance management, and organizational learning. The course emphasizes contemporary HRD strategies, tools, and evaluation methods applicable across various industries. Teaching methods include interactive lectures, case analyses, and group projects. By the end of the course, students will be able to design and assess HRD interventions that align with organizational goals.

Compulsory

Compulsory

When writing Course Learning Outcomes (CLOs), each outcome must align with the relevant Program Learning Outcomes (PLOs) and **explicitly reflect the sustainability competencies** selected, such as System Thinking, Anticipatory Thinking, Normative, Strategic Thinking, Collaboration, Critical Thinking, Self-Awareness, Integrated Problem Solving, Adaptability, and Flexibility, in accordance with the Malaysian Qualifications Framework (MQF) clusters of learning outcomes. CLOs must be formulated using specific and measurable action verbs from Bloom's Taxonomy at the appropriate cognitive, psychomotor, or affective level, and must follow the SMART principle—specific, measurable, achievable, relevant, and time-bound—so that they are observable and assessable within the course duration. Each course should ideally have three to five CLOs, avoiding vague or non-measurable verbs, and ensuring constructive alignment with both teaching and learning activities as well as assessment strategies.

Compulsory

Compulsory

Ensure that each Course Learning Outcome (CLO) in the Student Learning Time (SLT) table aligns with the corresponding weekly topics and activities. Weekly topics must match the course content and teaching schedule. Indicate which CLO(s) are addressed each week, ensuring a progression that reflects the course's learning goals. This mapping should be consistent with previously stated CLOs and supported by appropriate teaching methods and assessments.

Please ensure that all assessment information is consistent across the Mapping of Course Learning Outcomes (CLOs) to Programme Learning Outcomes (PLOs), Teaching and Learning (T&L) methods, and Assessment methods. This includes the tables for continuous and final assessments. Each CLO should be clearly linked to the corresponding assessment methods and their breakdown. Specify the types of assessments used, such as quizzes, assignments, projects, presentations, and final exams. All assessments must be measurable, reflect the cognitive level of the CLOs, and maintain an appropriate balance between continuous and final components.

30-79% = PTG

Please include at least two to three recent references (published within the last five years) in the course information to support the relevance and currency of the course content. Ensure that the references are credible, up-to-date, and properly cited using a consistent referencing style (e.g., APA).

Compulsory.
Signatures should be handwritten, not typed or styled with fonts.